

A Dehazing Network and Self-Supervised Transfer Learning Method in Highway Surveillance Scenes

Zhiyong Peng¹, Yuxiang Chen, Jiang Du, and Yulong Qiao², *Member, IEEE*

Abstract—This paper focuses on the application of image dehaze algorithms in highway scenarios, proposing a novel dehaze algorithm and a self-supervised transfer learning method for practical highway surveillance applications. The new lightweight dehazing network with the pyramid network structure is designed by combining the information multi-distillation network (IMDN), the channel and pixel attention module. In the deployed highway monitoring application, the self-supervised transfer learning method by proposed by integrating the pre-trained dehazing model with a dynamic target detection network. Through multiple alternating learning processes, the dehazing model continuously transfer and suitable for the current real-world application scenarios. The proposed algorithm is rigorously tested on an RTX 3090 GPU by using several public standard datasets and real-world highway datasets. The results demonstrate that the new algorithm outperforms state-of-the-art algorithms, achieving significantly higher Peak Signal-to-Noise Ratio (PSNR) and structural similarity (SSIM) on the public datasets. Furthermore, the visual quality of the dehazed images from new algorithm after transfer learning is markedly superior compared to other algorithms in the real-world highway scenarios. In terms of speed, the new algorithm exhibits faster inference speed than other comparative algorithms, achieving a frame rate 25 frames per second (FPS) for the 1920×1080 real video. On the 4KID dataset, the inference speed can reach 26ms.

Index Terms—Image dehazing, deep learning, self-supervised transfer learning.

I. INTRODUCTION

THIS dehaze technology significantly enhances the clarity and detail of traffic images captured along highways and long-distance video surveillance footage under adverse weather conditions such as fog and haze. It accurately restores

critical information including road vehicles, pedestrians, traffic signs, and lane markings, thereby greatly improving the effectiveness and accuracy of highway traffic monitoring systems. This provides reliable data support for core tasks such as traffic management, accident early warning, and incident response [1]. Leveraging their robust feature extraction capabilities, deep neural networks achieve remarkable results in highway traffic image dehaze by precisely removing fog interference while preserving critical traffic elements. Through continuous iteration of deep learning algorithms and ongoing upgrades to highway traffic monitoring hardware, their synergy provides innovative solutions and stable platform support for image dehaze in highway traffic scenarios [2]. Consequently, deep learning-based image dehaze technology tailored for highway traffic scenarios represents a prominent research topic aligned with the demands of intelligent transportation development.

Cai et al. [3] introduced DehazeNet, a CNN aimed at image dehazing, which demonstrated promising dehazing effects. However, the atmospheric scattering model employed in DehazeNet is a simplified representation of reality, resulting in suboptimal dehazing performance. In response, many researchers have begun to explore fully end-to-end deep learning networks, such as GFN [4], GCANet [5], DANet [6], etc. [7] [8] [9]. With the rise of diffusion models, several deep learning networks based on diffusion model designs have emerged, demonstrating exceptional dehaze performance, such as FCD [10] and Diff-Dehazer [11]. Concurrently, in recent years, dehaze networks tailored for real-world scenarios have also seen significant development, including DNMGDT [12], D4 + [13], and DehazeSB [14]. These algorithms have shown significant improvements in peak signal-to-noise ratio (PSNR), but they often come with a high computational cost, making real-time dehazing challenging. Additionally, these methods tend to struggle with color reproduction and the effective removal of dense fog in complex real-world scenarios, limiting their practical applicability.

This paper presents a lightweight dehazing network based on the Laplace pyramid structure, which integrates an attention mechanism and the Information Multi-Distillation Network (IMDN). At the same time, we present a self-supervised transfer learning method for highway monitoring applications, enabling the model to continuously adapt and enhance its dehazing capabilities for real-time scenarios. The main contributions of this paper include:

- a) To achieve an optimal balance between speed and quality, the paper presents a novel lightweight deep learning dehazing network based on Laplacian pyramid structure,

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Zhiyong Peng, Yuxiang Chen, and Jiang Du are with Guangxi Key Laboratory of Optoelectronic Information Processing, School of Optoelectronic Engineering, Guilin University of Electronic Technology, Guilin 541004, China (e-mail: pzy@guet.edu.cn; cyx@mails.guet.edu.cn; 731196971@qq.com).

Yulong Qiao is with the School of Information and Communication Engineering, Harbin Engineering University, Harbin 150001, China (e-mail: qiaoyulong@hrbeu.edu.cn).

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combined with IMDN, as well as channel and pixel attention modules. The network is capable of dehazing and displaying videos with a resolution of 1920×1080 in real time.

- b) To enhance dehaze performance in real-world highway surveillance scenarios, this paper proposes a novel self-supervised transfer learning approach. By integrating a dehaze model with a dynamic object detection network tailored for highways, the method achieves sustained improvement in dehaze capabilities through multiple rounds of alternating iterative learning. It enables self-iterative algorithmic upgrades to adapt to dynamic road conditions and environmental changes in real-time highway monitoring.
- c) This paper proposes a method that combines atmospheric scattering models with depth maps to effectively enhance the realism of network training data, enabling the generation of simulated fog images tailored to highway traffic scenarios.

The remainder of the paper is as follows: Section II reviews relevant literature and related studies. Section III details the proposed dehazing algorithm. Section IV analyzes and discusses the experimental results obtained from various datasets. Finally, Section V concludes the paper with final remarks.

II. RELATED WORK

A. Laplace Pyramid

The Laplacian pyramid is a multiscale representation method used to describe image features at different scales, and it is widely applied due to its excellent multiscale fusion capabilities. For instance, Zhang et al. [15] proposed an end-to-end single-image dehaze method that combines an encoder-decoder architecture with a pyramid pooling module to estimate the transmission map, enabling single-image dehaze processing. Xiao et al. [16] achieved dehaze for ultra-high-definition images by integrating a pyramid structure with Taylor's theorem. Chen et al. [17] addressed the issue of image forgery by proposing a Signal-Noise Separation (SNIS)-based network. This approach first performs signal-noise separation, followed by a multi-scale feature learning module to learn global features. Inspired by these approaches and aiming at the specific demand of haze removal for highway scene images, we utilize the pyramid structure to design a dedicated image dehaze network for highway scenarios.

B. Transfer Learning

In recent years, deep learning-based methods have achieved significant progress in image dehaze. However, most existing approaches train dehaze models on synthetic foggy images. Due to domain shift, these models perform poorly when applied to real foggy images. Transfer learning leverages knowledge from the source domain to aid target domain learning, offering a solution to issues such as poor target domain adaptability, insufficient data, and high annotation costs [18]. It is frequently applied to various image restoration tasks. For instance, Wei et al. [19] proposed a semi-supervised transfer learning approach for image de-raining, extending the

de-rain network from adapting only to synthetic rain in the source domain to handling real rainfall in the target domain. Chi et al. [20] introduced a self-supervised meta-assisted learning method to further optimize pre-trained models, enhancing their performance in addressing motion blur in dynamic scenes. Inspired by these approaches, we explored applying self-supervised transfer learning tailored to real highway surveillance scenarios to enhance the performance of dehaze networks on authentic foggy images.

III. METHODS

A. Lightweight Dehazing Network Design Based on Pyramid Structure

1) *Overall Structure of the Dehazing Network:* In this paper, the dehazing algorithm is designed based on Laplace pyramid structure to achieve an optimal balance between speed and quality. This approach employs Gaussian smoothing and downsampling to separate the image into distinct spatial bands. Feature extraction is conducted for all band to capture features and detailed information across various levels. Ultimately, the features extracted from these distinct bands are fused to produce the final result. The network contains the IMDN, channel and pixel attention modules. The overall structure of the network is presented in Figure 1.

In the network structure, images undergo Gaussian smoothing and downsampling to capture low-frequency information. This low-frequency data is then upsampled, and high-frequency information is subtracted to isolate high-frequency edge details. The final image is divided to extract high-frequency and low-frequency features: The low-frequency features reflect the main features such as color and illumination, while the high-frequency features reflect the edge information of the image.

After splitting, deep feature extraction can be performed with less computation for color, illumination, etc. Due to the high structural similarity of high-frequency information at different scales, this paper upsamples low-frequency features to assist in extracting high-frequency features. The importance of different pixel features of the same channel and the importance of features of different channels are learned by pixel attention and channel attention respectively, and finally the outputs of different layers are fused together to get the final output.

2) *Improved Information Multi-Distillation Network (IMDN):* The information multi-distillation feature extraction block was first proposed by Hui et al. [21], demonstrating improved reconstruction quality and inference speed in image super-resolution tasks. In this paper, we intend to apply the information multi-distillation extraction block in conjunction with a pyramid structure for the task of dehazing. Specifically, we use five information multi-distillation blocks (IMDB) along with convolutional layers to create the final IMDN. The structure of the IMDN is illustrated in Figure 2.

Since the low-frequency information contains a large amount of color, brightness and other information, so this paper hopes to retain the feature information obtained from each layer of the network as much as possible, this paper designs the IMDN through the lightweight feature extraction block IMDB combined with the residual idea of retaining as

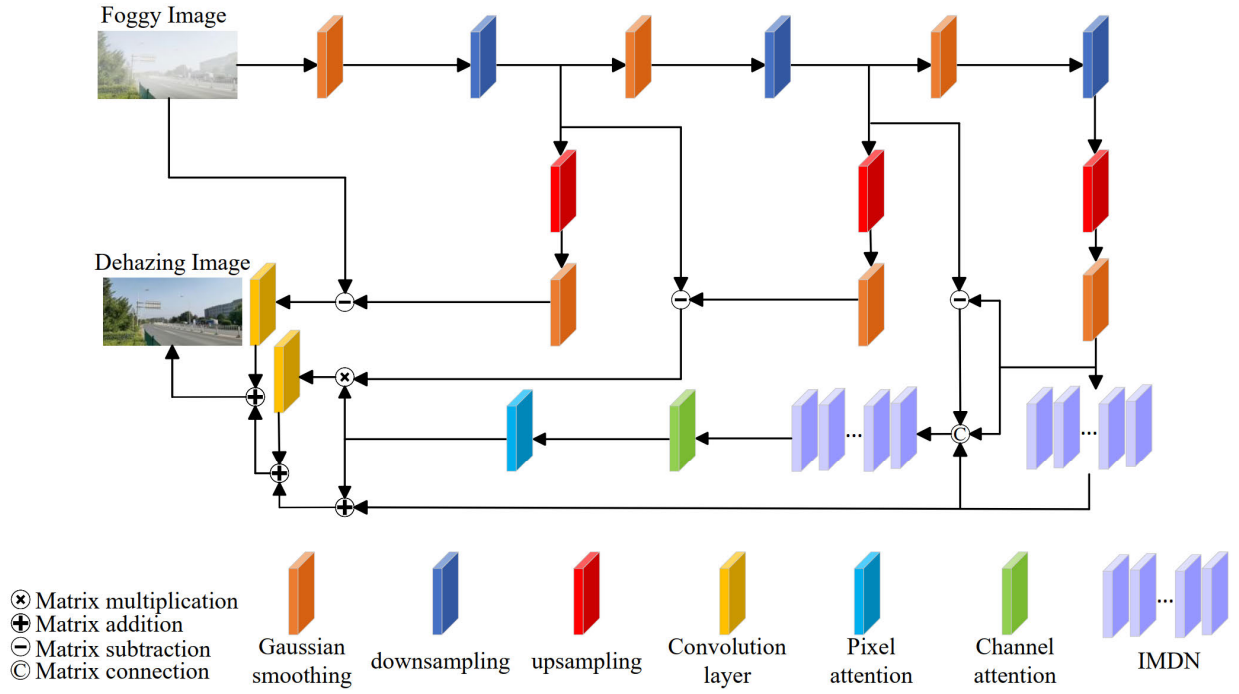


Fig. 1. Overall structure of dehazing network based on pyramid structure.

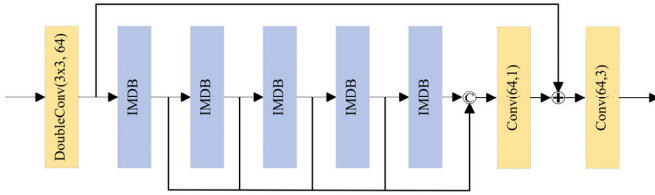


Fig. 2. The structure of the IMDN.

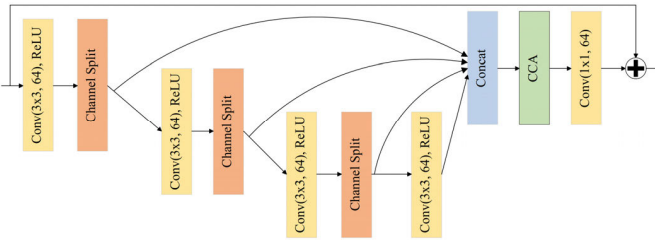


Fig. 3. The structure of the IMDB.

much as possible the texture information at the same time to learn more features. The DoubleConv layer refers to the process of splitting the number of channels to 64 into two convolutional operations so that the two 32-channel features are merged into a 64-channel can reduce the inference time, and the residuals idea will be used to connect the modules as a pyramid of low-frequency information feature extraction backbone network. The structure of the IMDB is illustrated in Figure 3.

The IMDB is built by using residual thoughts, and gradually refines the features through continuous channel splitting. All features undergo contrastive perceptual channel attention (CCA) layer and 1×1 convolution to reduce the number of feature channels. Where the contrast perceptual channel attention (CCA) is displayed in Figure 4.

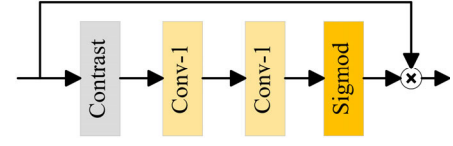


Fig. 4. Contrast-Aware Channel Attention (CCA).

In this case, the progressive refinement module splits the channels of the previous layer of features, some of which are retained as refined features, while others go to the next computational unit. This preserves a large number of features and reduces the computational load in each layer. After joining all the information and feeding it into CCA, the common use of pooling operation usually improves the final PSNR value, but leaves the image missing the image texture, edge information, etc. So the use of CCA aims to replace the global average pooling with the sum of the mean and the standard deviation, retaining more detailed information can help the final result obtained by fusing with pyramidal high frequency features is richer in texture information, assuming that the input $X = [x_1, \dots, x_c, \dots, x_C]$ has C layers of features with the dimensions of the size of the $H \times W$ size, and the specific calculation is shown in equation (1).

$$\begin{aligned}
 Z_c &= H_{GC}(x_c) \\
 &= \sqrt{\frac{1}{HW} \sum_{(i,j) \in x_c} \left(x_c^{i,j} - \frac{1}{HW} \sum_{(i,j) \in x_c} x_c^{i,j} \right)^2} \\
 &\quad + \frac{1}{HW} \sum_{(i,j) \in x_c} x_c^{i,j}
 \end{aligned} \quad (1)$$

where Z_c represents the c -th element of the output and H_{GC} represents the global contrast information evaluation function. Based on the contrast perception, the pixel-level contrast inside the channel is considered, and the channel with higher contrast

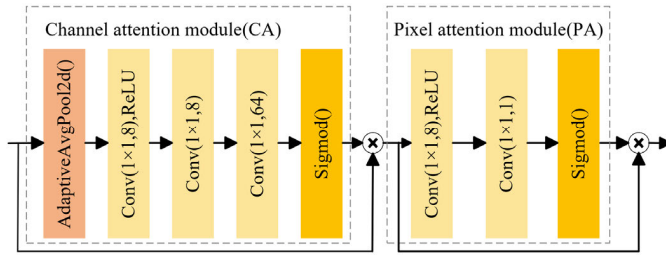


Fig. 5. Combined structure diagram of channel and pixel attention modules.

inside the channel is considered more important to retain important texture information.

3) *Attention Module Combining Channel Attention and Pixel Attention:* In this paper, both the channel attention mechanism and the pixel attention mechanism [6] [22] [23] are used to help the network further improve its performance. In this paper, we incorporate two attention modules (channel attention module (CA) and pixel attention module (PA)) to guide the network in focusing on essential feature information. The effectiveness of these modules is demonstrated through experimental results. The specific structure of the attention modules is illustrated in Figure 5.

The channel attention module (CA) inputs feature $f = [f_1, f_2, \dots, f_c]$, firstly, all positional eigenvalues $v = [v_1, v_2, \dots, v_c]$ of feature f are obtained by average pooling:

$$v = \frac{1}{HW} \sum_{i=1}^W \sum_{j=1}^H f(i, j) \quad (2)$$

where, W and H are the width and height of the feature map. i, j denote the coordinate positions. The feature v under-goes three convolution operations with a 1×1 convolution kernel. The sigmoid function is applied to retain the important features (the channel weight parameters).

The principle of pixel attention module (PA) is similar to the CA, except that the input feature f is changed to 1 channel through the convolution. The pixel attention value is obtained through the Sigmoid activation function. In the end, the result of the pixel attention is multiplied with the input feature f to preserve important pixel features.

By applying the attention module to features, the network enhances its ability to focus on dense fog and critical scene information. Simultaneously, the attention output is upsampled and integrated with high-frequency features, enabling the upper layers of the pyramid network to better capture high-frequency detail. This approach significantly improves the overall dehazing performance of network and enhances the restoration of texture details.

B. Self-Supervised Transfer Learning for Real Highway Surveillance Scenarios

This article employs a transfer learning approach to transfer the parameters learned from a synthetic dataset to real-world highway monitoring scenarios. This method enables the continuous learning of dehazing tasks relevant to current highway conditions. We utilize clear weather images captured in real surveillance environments as label images, while foggy degraded images in real surveillance scenes as training data.

But for the transfer learning there is a key problem, that is, how to solve the influence of a large number of possible people, cars and other moving targets for training results. This paper designs a dynamic target detection network to help transfer learning to remove the influence of dynamic targets. The transfer learning ideas designed in this paper are theoretically self-supervised and applicable to all kinds of scenes, as a verification of this paper, we choose the outdoor highway to illustrate and introduce.

1) *The Overall Process of Transfer Learning:* In the transfer learning process of the pyramid structured dehazing network, this article also needs to train the dynamic target detection network, aiming to remove the influence of dynamic targets such as vehicles and people on the training results on highways. The dynamic target detection network and the dehazing network are trained through continuous alternating iterations. During the final training process of the dehazing network, the parameters of the dynamic target detection network are not updated and only verified. Conversely, during the training phase of the dynamic target detection network, the parameters of the dehazing network are not updated. The dehazing network and the dynamic target detection network that are suitable for the current scene are obtained through continuous alternating iterations in the large amount of real data collected from the current camera. The specific process is displayed in Figure 6.

As illustrated in Figure 6, the effect of dynamic targets on network training is removed by multiplying the output of the dynamic target detection network inversely with the output matrix of the dehazing network.

In the real scene, the dynamic target is not only cars and people, but also includes lighting, natural airflow and other factors that change the surrounding environment and affect the network transfer learning, and make it difficult for the transfer learning process to converge. This paper does not update and retrain the entire dehazing network during the transfer learning process. Instead, it focused on adjusting output layer parameters to avoid losing control of transfer learning training in practical applications. The specific transfer of the dehazing network is displayed in Figure 7. During transfer learning training, only the output part of each layer on the pyramid is trained and updated, that is, the convolutional layer and two attention modules in the output part are fine-tuned for parameter training.

2) *Dynamic Target Detection Network:* To mitigate the interference of dynamic target (such as moving vehicles) on transfer learning performance in highway surveillance scenarios, designing a dedicated dynamic target detection network is crucial. The network constructed in this study performs targeted processing on highway surveillance images, outputting a binary representation of dynamic target (where dynamic objects are labeled as “white” and highway background areas as “black”). The specific architecture of this dynamic target detection network is illustrated in Figure 8.

In this paper, the dynamic target detection network refers to the feature extraction idea of YOLOv5 [24], but the final goal of this paper is different from the YOLOv5 target detection algorithm, and the final output of this paper’s dynamic target

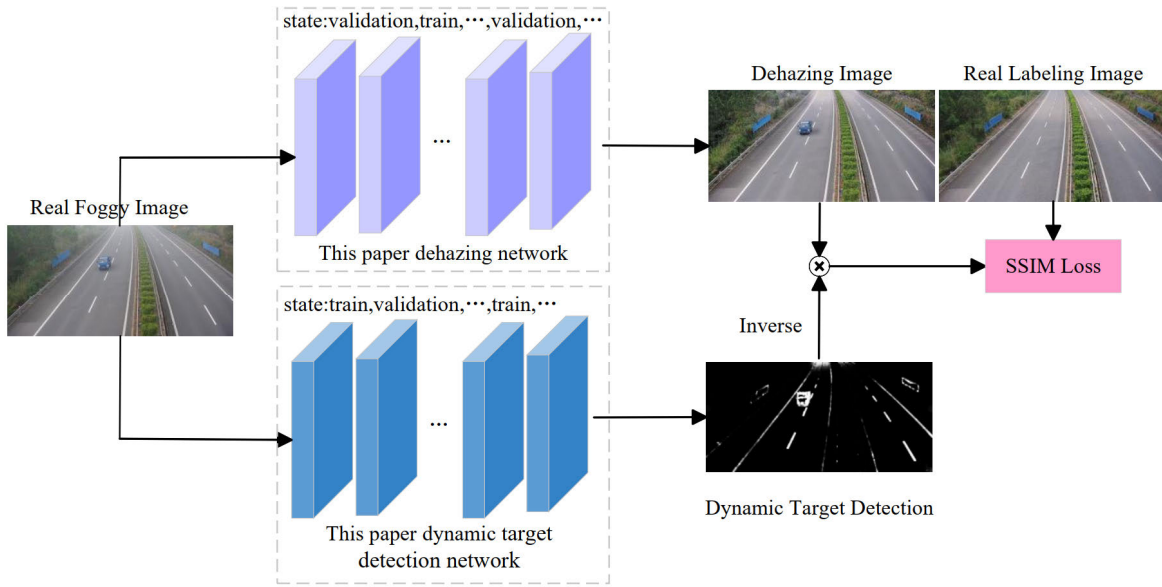


Fig. 6. Overall structure of image dehazing transfer learning.

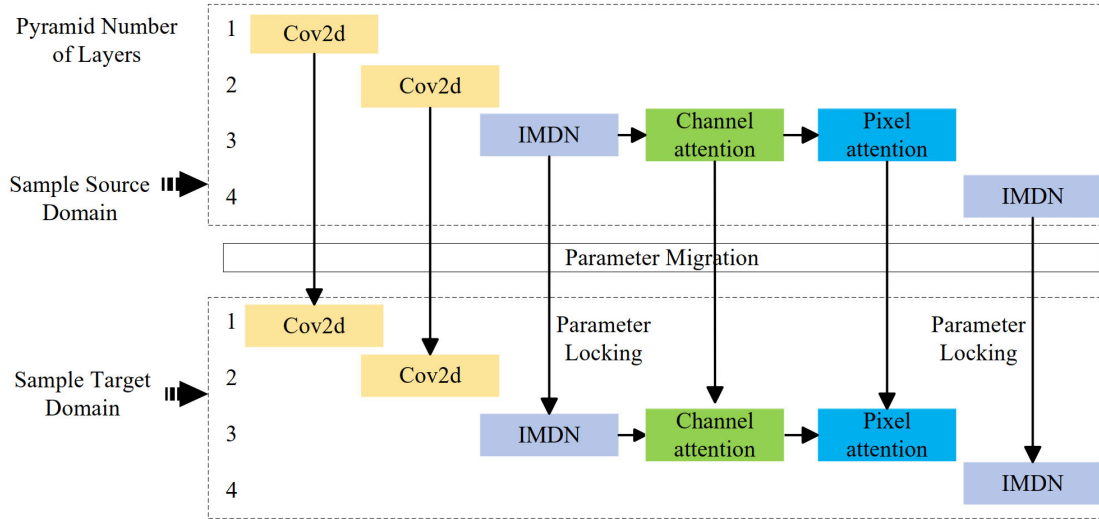


Fig. 7. Migration of image dehazing parameters in transfer learning.

detection network is a single-channel binarized matrix. In this paper, the input is a 3-channel (RGB) color image, which undergoes a series of convolution operations and local residual structures to extract 64 feature channels. Finally, the feature channels is diminished to 1, and the Sigmoid activation function is applied to normalize the network output between 0 and 1, representing the probability of identifying a dynamic target. This dynamic target probability is then compared to threshold value for getting binarized image. Additionally, the use of max pooling during the convolution process effectively mitigates overfitting, expands the receptive field, and enhances the network's generalization capability.

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C. Generation of Training Datasets

The algorithm structure for generating foggy images in the training dataset is shown in Figure 9. Obtain depth information through depth estimation networks. Combining depth information with atmospheric scattering models to generate foggy images corresponding to label images through random brightness changes.

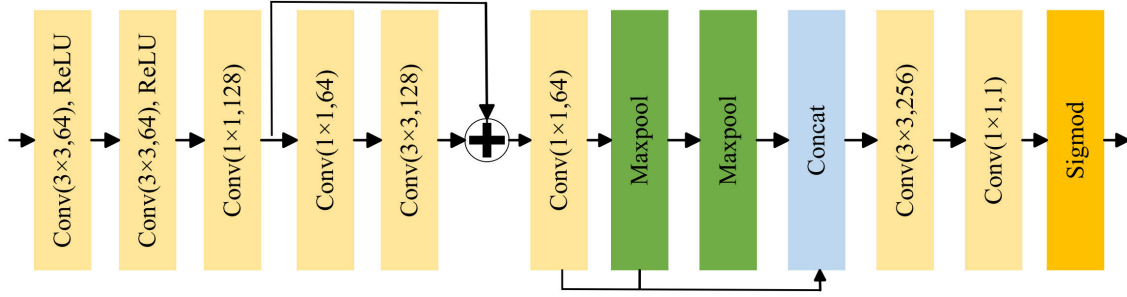


Fig. 8. Dynamic target detection network structure.

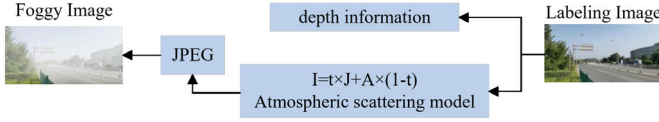


Fig. 9. Algorithm structure for generating foggy images.

The depth information is the monocular image depth data estimated by the Depth Anything V2 model [25]. This model is constructed based on a pre-trained DINOv2 encoder and a DPT depth decoder, enabling end-to-end processing of input images. By performing image preprocessing, feature extraction, and pixel-level depth computation, it ultimately outputs precise depth maps.

The atmospheric scattering model can be used to explain the scattering phenomena caused by the interaction between light and particles such as air molecules and aerosols in the atmosphere [26]. The foggy degradation formula of the atmospheric scattering model is shown in equation (3)

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (3)$$

where $I(x)$ is a foggy image, $J(x)$ is a fog-free image, A is the value of atmospheric light, and $t(x)$ is the transmittance.

When synthesizing foggy images, this paper considers various degradation factors based on traditional atmospheric scattering models, such as, insufficient light in foggy images, uneven distribution of fog related to depth, and noise, thereby reducing the difference between the generated foggy images and real foggy images. The fog degradation formula optimized based on atmospheric scattering model is shown in (4).

$$I(x) = \text{JPEG}\left(P(J(x)^\gamma + N), e^{\beta d(x)}, A\right) \quad (4)$$

where $\text{JPEG}()$ represents JPEG compression of the final result. In response to changes in lighting conditions, a brightness adjustment factor γ was designed, and the value of γ was randomly selected within $[1.5, 3.0]$ for each foggy image. N is a Gaussian noise. γ , N these two parameters simulate the situation of insufficient light that often occurs in foggy weather.

The concentration of real scene fog must be non-uniformly distributed, the depth estimation algorithm [27] is used to estimate the depth map $d(x)$, and $\beta \in [0.3, 1.5]$ is used to adjust the smoke concentration.

IV. RESULTS

This paper compares the performance of the new algorithm with classical algorithms across three public datasets.

Additionally, we evaluated the changes before and after applying self-supervised transfer learning in real highway scenes and compared the dehazing effects of the new approach with classical algorithms.

A. Datasets

4KID: This is a 4K (3840×2160) resolution dataset with 13,000 outdoor foggy-free images of different scenes. We generate foggy maps for training, validation, and comparison testing using the methodology in Section III. METHODS, C. Generation of training datasets of this paper.

1) I-Haze: This is an indoor foggy dataset which consists of different indoor scenes. The fuzzy images in this dataset are generated using a professional fogging machine. This dataset is mainly used for testing in the paper.

2) O-Haze: This is an outdoor haze dataset, which consists of different outdoor scenes. The blurred images in this dataset are generated using a professional fogging machine. This dataset is mainly used for validation in the paper.

3) Real Highway Dataset: This is a dataset composed of real highway surveillance images, including foggy images and clear-day images without moving objects. The foggy images are used as training images, and the clear-day images without moving objects are used as label images. This dataset is mainly used in the transfer learning section of the paper.

Fog free real highway dataset: This consists of highway surveillance images without fog and with dynamic targets. We have labeled the dynamic targets within these images. This dataset is mainly used for the pre-training of dynamic target detection networks.

B. Experimental Settings

This paper uses a generated foggy dataset to train an initial dehazing network model, which is then utilized as a pre-trained model for transfer learning in dehazing tasks. This paper uses the Fog free real highway dataset to train an initial dynamic target detection network model, which is then used as the pre-training model for the dynamic target detection network. The training process detailed in this paper comprises three core stages: the training of the dehazing network, the training of the dynamic target detection network, and the transfer learning phase.

1) Dehazing Network Model Training: During the training process of dehazing network, the Adam optimizer is used for updating the network parameters. The learning rate is

TABLE I

CONTRASTING THE EFFECTS OF VARIOUS FOG GENERATION PROCESSES

Method	MOS
Atmospheric scattering model	2.1
Atmospheric scattering model combined with depth map (γ)	1.7
Atmospheric scattering model combined with depth map (No γ)	3.8

set to 0.0002. the exponential decay rate of the first-order moment estimation is 0.9, and the exponential decay rate of the second-order moment estimation is 0.999, and the learning rate is adjusted along with the training process by the cosine annealing algorithm. The number of iterations is 80 epochs and a batch is 4. The loss function is charbonnier [28] and the formula is shown in (5).

$$loss = \sqrt{x^2 + \varepsilon^2} \quad (5)$$

where, x is the difference between the output map and the labeled map. ε is a constant, and it is set to 0.001 in this paper, The loss is a differentiable variant of the L_1 paradigm [29], while compared to the mean square error loss can help the network to better handel with the outliers and improve the performance of the network.

2) *Dynamic Target Detection Network Training*: During the training of the dynamic target detection network, the Adam optimizer is adopted to update the network parameters with the following configurations: a learning rate of 0.0001, an exponential decay rate of 0.9 for the first-order moment estimation, 0.999 for the second-order moment estimation, a training iteration of 10 epochs, a batch size of 8, and the SSIM as the loss function.

3) *Self-Supervised Transfer Learning*: After acquiring the pre-trained dehazing network and dynamic target detection network models, transfer learning is employed to further enhance the dehazing performance in practical monitoring scenarios. Specifically, in the actual highway monitoring applications, the monitoring system can automatically collect images as labels during sunny days, while degraded images are collected in foggy environments. For self-supervised transfer learning, highway foggy-day images are used as the input, while highway sunny-day images are used as the label images. After being processed by the de-fogging network and the dynamic target detection network, the resulting images are then compared with the label images using the static point average SSIM loss as the loss function. The process of transfer learning is shown in Figure 6.

The transfer learning process proceeds as follows: First, the pre-trained dehazing network is frozen, and the pre-trained dynamic target detection network is fine-tuned until the SSIM loss stabilizes. Subsequently, the dynamic object detection network is frozen, while the dehaze network was trained based on the pre-trained model. At this stage, only the convolutional layer in the output layer and the two attention modules underwent fine-tuning until the SSIM loss reached a stable state. These two steps are cyclically repeated until the

SSIM loss no longer exhibits significant changes following each switch between the two networks.

By leveraging real highway scene image data, the network performs continuous cyclic self-supervised transfer learning. This approach effectively improves the output quality of the dehazing network while continuously optimizing the overall network performance for practical highway monitoring applications.

C. Evaluation Indicators

We use PSNR and SSIM, which are commonly used in the area of dehazing algorithms, to assess the performance of image dehazing [16]. Additionally, we use Mean Opinion Score (MOS) to assess whether the synthesized fogged dataset can accurately reflect the real situation.

MOS, which represents the authenticity of simulated foggy image, the lowest score is 1 and the highest score is 5. According to reference [30], the authenticity value (AV) of each simulated image in the dataset is obtained by calculating it using (6). Then, the SRCC and PLCC values of the MOSs are obtained by calculating the AV of the entire dataset. These high correlation values indicate that the prediction of realism of the simulated fog image is in high agreement with the subjective assessment results.

$$D = \sqrt{(u_1 - u_3)^T \left(\frac{\sum 1 + \sum 3}{2} \right)^{-1} (u_1 - u_3)} + \sqrt{(u_2 - u_4)^T \left(\frac{\sum 2 + \sum 4}{2} \right)^{-1} (u_2 - u_4)} \quad (6)$$

where $\mu_1, \mu_3, \mu_2, \mu_4$ and $\sum 1, \sum 2, \sum 3, \sum 4$ are the covariance matrix and mean vector of the simulated image MVG model and the natural MVG model.

D. Test Results and Comparative Analysis

For the experiments in this paper, the algorithms were trained and tested on a computer with Ubuntu 20.04, a CPU of i9-10900K and a GPU of NVIDIA RTX3090.

1) *Evaluation of the Quality of Simulated Foggy Image Generation*: In this work, we simulate a realistic foggy scene by combining an atmospheric scattering model with a depth map. The generated foggy image from new approach is closer to the real foggy situation than using atmospheric scattering model alone. The generated foggy images from the 4KID dataset are used to calculate MOS, the data are illustrated in Table I.

In Table I, γ represents the introduction of random brightness adjustment, and No γ represents no introduction of random brightness adjustment. The foggy image generation method combining depth map without introducing random brightness far exceeds the MOS value of using atmospheric scattering model alone. After the introduction of random brightness, the change in brightness will make the overall image too dark or overexposed, so the MOS value becomes low. To achieve a balance between the generalization ability of the work with the authenticity of the generated foggy images, 20% of the training set consists of images with randomized

TABLE II
COMPARISON TEST DATA WITH CLASSICAL NETWORKS (PSNR/SSIM/TIME)

datasets	metrics	GCA[5]	PFF[31]	MSBDN[32]	FCD[10]	DNMGDT[12]	DehazeSB[14]	ours
4KID	PSNR	19.26	21.52	22.04	21.56	20.21	19.89	26.09
	SSIM	0.834	0.735	0.841	0.853	0.813	0.821	0.923
	Time	95ms	38ms	-	2061ms	188ms	253ms	26ms
I-HAZE	PSNR	14.57	16.41	19.93	17.56	17.95	17.48	18.82
	SSIM	0.719	0.742	0.821	0.826	0.832	0.745	0.792
	Time	101ms	42ms	-	2062ms	191ms	251ms	37ms
O-HAZE	PSNR	16.05	18.93	19.36	17.37	18.21	17.36	20.77
	SSIM	0.743	0.780	0.798	0.807	0.796	0.737	0.816
	Time	102ms	45ms	-	2066ms	189ms	253ms	39ms

TABLE III
PROCESSING TIME FOR DIFFERENT RESOLUTION IMAGES

Datasets	inference	pre- and post-processing	total
4KID(4K)	26ms	51ms	77ms
Highway dataset in this paper (1080×1920)	19ms	20ms	39ms

TABLE IV
COMPARISON OF THE EFFECTS OF INTRODUCING THE ATTENTION MODULE

Experiments	I-Haze		O-Haze	
	PSNR	SSIM	PSNR	SSIM
No attention module + No IMDN	13.31	0.643	13.67	0.658
No attention module + IMDN	14.36	0.692	15.11	0.711
CA + IMDN	17.89	0.760	18.00	0.780
PA + IMDN	16.64	0.714	16.75	0.731
CA + PA + IMDN(Ours)	18.82	0.792	20.77	0.816

brightness levels. Overall, the presented method is more in line with the real fog situation, and take into account the different light conditions to improve the generalization ability of the network.

2) *Comparison of the Effect of Different Algorithms for Public Datasets:* The experiments compare new algorithm with the classical dehazing algorithms GCA [5], PFF [31], MSBDN [32], FCD [10], DNMGDT [12], DehazeSB [14]. The result data about peak signal-to-noise ratio (PSNR) of the images, structural similarity (SSIM) and the inference times of the different methods are shown in Table II. In this paper, the common public test set 4KID, I-HAZE and O-HAZE are used to test the deference algorithms. The proposed algorithm achieves good PSNR and SSIM on public datasets, and the speed of the new algorithm is the fastest. The PSNR and SSIM values for the new algorithm on the 4KID and O-HAZE dataset is much higher than state-of-the-art algorithms. The PSNR of new algorithm has an improvement of 4.05dB over the MSBDN. I-HAZE is an indoor public dataset, this paper

focuses on outdoor scenes, so most of the training data are indoor scenes, which lead to the PSNR and SSIM of new algorithm is suboptimal on dataset I-HAZE. Additionally, the average time of model inference for different algorithms are tested, this paper is the fastest on all dataset. In the Table II, “-” means no speed comparison. Because the MSBDN network requires the input image must be a multiple of 4, which leads to many images do not meet the setting of tested directly, so MSBDN does not have time data. It is evident that the algorithm proposed in this paper ensures better PSNR and SSIM data while the network inference speed exceeds the comparison algorithm on public outdoor dataset.

In Table II compares the average inference speed of the models of various types of algorithms on the dataset, but the inference speed does not represent the frame rate that can be achieved in the actual application. Different algorithms have different pre- and post-processing, then the pre- and post-processing often require a significant amount of time. The pre- and post-processing in this paper contains the transmission

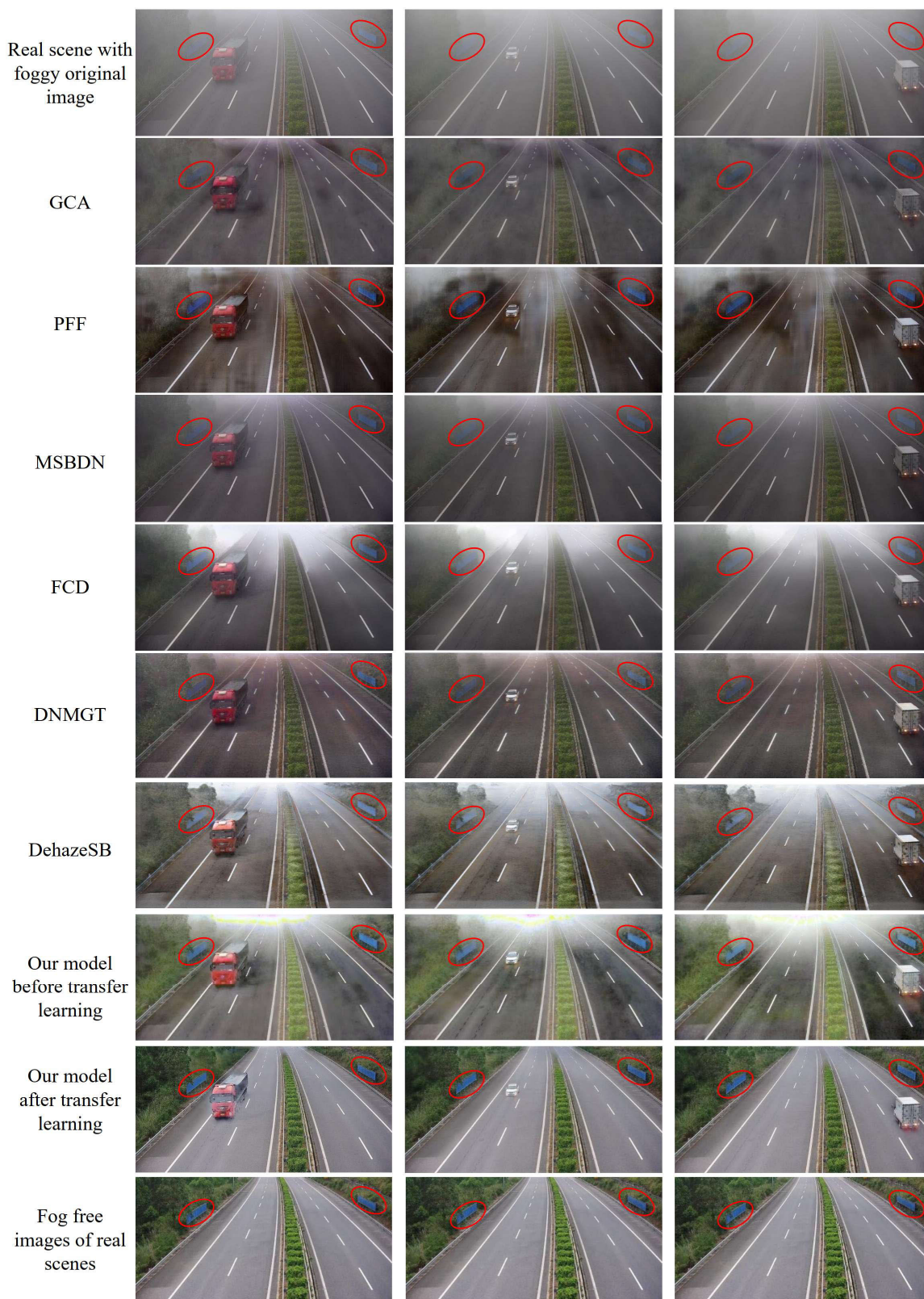


Fig. 10. Comparison of the effects of transfer learning on dehazing in real outdoor highway.

of data between the CPU and GPU, the deformation of the matrix, the reading of the image and the image display in real time, etc.

According to the actual situation, the speed of this paper in practical applications from image input to output is displayed in Table III.

As shown in Table III, with the high resolution, the model inference speed, pre- and post- processing will consume a lot of time. In summary, the real-time dehazing of 1920×1080 resolution images in the real application of this paper can reach 25 FPS, with an average processing time of 39ms.

3) *Self-Supervised Transfer Learning Testing in Highway Monitoring Applications*: This article utilizes a dataset comprising 5,000 real foggy images and 5,000 real sunny images sourced from highway monitoring applications on the Guigang-Hepu Highway within the Guangxi Zhuang Autonomous Region of China to validate the transfer learning effectiveness of our algorithm. The dehazing effect after transfer learning in a real outdoor scene of the Guigang-Hepu Highway is illustrated in Figure 10. Figure 10 shows that the two blue roadside signs (marked in red) are located approximately 60 meters from the surveillance camera. It is evident that the dehaze algorithm presented in this paper effectively enhances the clarity and visibility of surveillance images in foggy conditions, providing high-quality image data for highway traffic monitoring. To better illustrate the difference between the dehazing effect and the actual image, this article includes a clear and fog-free scene of the Guigang-Hepu Highway in the final line of the test results. From the actual test results on the Guigang-Hepu Highway, it is evident that the dehazing effect of our algorithm before transfer learning is better than that of the comparison algorithm. Furthermore, our algorithm has achieved significant improvements in dehazing performance through self-supervised transfer learning tailored to monitoring scenes, particularly in effectively removing distant dense fog on the Guigang-Hepu Highway. After transfer learning, the results of the tests indicate that our algorithm surpasses other algorithms.

E. Ablation Experiment

This section validates the effectiveness of the two feature attention modules and IMDN introduced into the pyramid network architecture using the I-Haze and O-Haze datasets. Based on the pyramid network architecture, we compare the performance differences between network models incorporating attention modules and IMDN during feature extraction. The experiments encompassed the following configurations: no attention module and no IMDN; no attention module and IMDN; channel attention module (CA) and IMDN; pixel attention module (PA) and IMDN; and a combination of channel attention module, pixel attention module, and IMDN (CA+PA+IMDN). The specific comparison results are presented in Table IV.

The ablation experiment demonstrates that incorporating the feature attention modules and IMDN significantly enhances the dehazing performance of the network. For the O-Haze dataset, the combination of CA and PA generated a maximum PSNR improvement of 20.77 dB, which is 7.1 dB higher than the performance of the no attention module and no IMDN. It has been demonstrated that guiding the network to emphasize essential information in both the channel and pixel dimensions significantly impacts the final training results. Ablation experiments demonstrate that incorporating IMDN and feature attention modules significantly enhances the network's dehaze

performance. On the O-Haze dataset, the combination of IMDN, CA, and PA achieves a maximum PSNR improvement of 20.77 dB, surpassing the performance without attention modules and IMDN by 7.1 dB. This experiment demonstrates that incorporating IMDN and attention modules substantially enhances the effectiveness of the proposed dehaze network.

V. CONCLUSION

This article proposes a lightweight image dehazing network based on Laplacian pyramid structure and a self-supervised transfer learning method in highway monitoring applications to address the problems of existing dehazing algorithms in practical applications. Self-supervised transfer learning method in highway monitoring applications involve integrating a pre-trained dehazing network with dynamic target detection network using a strategy of multiple alternating learning, so that the network can better adapt to dehazing in the current highway monitoring scenario. The dehazing network adopts a pyramid network structure, extracting complex features such as image color and brightness at the top of the pyramid, extracting image texture and structural features at the bottom of the pyramid, and finally fusing all features to obtain the dehazing image.

This paper also proposes a method for generating simulated foggy images by combining depth maps and atmospheric scattering models, which makes the generated foggy images closer to real foggy images and finally builds a training dataset. The lightweight dehazing network proposed in this paper significantly reduces computational requirements and is faster than other compared algorithms. It can achieve 25 FPS dehazing for 1920×1080 resolution images. In the public dataset 4KID, the PSNR index of the new dehazing algorithm is 26.09 dB, significantly surpassing that of other algorithms, it is 4.05 dB higher than the PSNR of MSBDN. Furthermore, our algorithm has achieved significant improvements in dehazing performance through self-supervised transfer learning tailored to monitoring scenes. After transfer learning, our algorithm significantly outperforms other traditional methods in outdoor highway scenes.

In the future, the proposed image dehaze algorithm and self-supervised transfer learning method can be further optimized and extended to broader application scenarios. When integrated into intelligent surveillance cameras or systems, cameras or systems can be designed to first capture foggy and clear-weather images on highways manually or automatically. Subsequently, self-supervised transfer learning can eliminate interference from moving objects and complete algorithmic iteration, continuously enhancing dehaze performance to adapt to changing road conditions and environments. This provides clear image inputs for advanced visual tasks such as object detection and traffic flow statistics, thereby strengthening the all-weather operational capabilities of Intelligent Transportation Systems (ITS).

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Zhiyong Peng received the Ph.D. degree in instruments science and technology from Guilin University of Electronic Technology, Guilin, China, in 2021.

He is currently a Professor of experimental science with the School of Optoelectronic Engineering, Guilin University of Electronic Technology. His research interests include computer vision, photogrammetry, and high-performance parallel image processing.



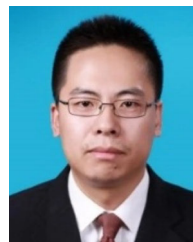
Yuxiang Chen received the B.S. degree from Guilin University of Electronic Technology, Guilin, China, in 2023, where he is currently pursuing the M.S. degree with the School of Optoelectronic Engineering.

His current research interests include image enhancement and deep learning.



Jiang Du received the M.S. degree from Guilin University of Electronic Technology, Guilin, China, in 2024.

His current research interests include image super segmentation, image enhancement, and deep learning.



Yulong Qiao (Member, IEEE) received the Ph.D. degree in instrument science and technology from Harbin Institute of Technology, Harbin, China, in 2006.

He is currently a Professor with the School of Information and Communication Engineering, Harbin Engineering University. His current research interests include image processing and signal representation and analysis.